

Deep Learning

Theoretical Exercises - Lesson 2 - Chapter 3

Exercises on the book "Deep Learning" written by Ian Goodfellow,
Yoshua Bengio, and Aaron Courville.

Exercises and solutions by T. Méndez and G. Schuster

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1 Exercises on Probability and Information Theory

1. Given a Bernoulli distributed random variable x with $P(x = 1) = \frac{3}{4}$ and $P(x = 0) = \frac{1}{4}$.
 - (a) Draw the probability mass function $P_x(x)$.
 - (b) Find the expected value of x , i.e. $\mathbb{E}_x[x]$.
 - (c) Find the variance of x , i.e. $\text{Var}_x(x)$.
2. Given a uniformly distributed continuous random variable x between -0.5 and 1 .
 - (a) Draw the probability density function $f_x(x)$.
 - (b) Find the expected value of x , i.e. $\mathbb{E}_x[x]$.
 - (c) Find the variance of x , i.e. $\text{Var}_x(x)$.
3. There are 5000 yellow and 100 red cabs in a city. In a hit-and-run accident, a witness saw a red cab. What is the probability that it was actually a red cab if witnesses in 95% of the cases state the car's colour correctly?
4. Given are the following sets of samples of two different classes c_1 and c_2 :
$$S_1 = \{5.3, 5.7, 6.1, 6.3, 6.6\},$$
$$S_2 = \{6.2, 6.5, 6.9, 7.7, 8.0, 8.3, 8.9\}.$$
 - (a) Estimate the mean (expected value) and the variance of both classes and sketch their probability density function assuming that the samples of both classes are normally distributed.
 - (b) Estimate the prior probabilities $P(y = c_i)$ of both classes.
 - (c) Assign the new samples $x_1 = 5$ and $x_2 = 7$ to the class with the highest posterior probability.
 - (d) Calculate the entropy $H(x | y)$ of x with $y = c_1$ and $y = c_2$ respectively.